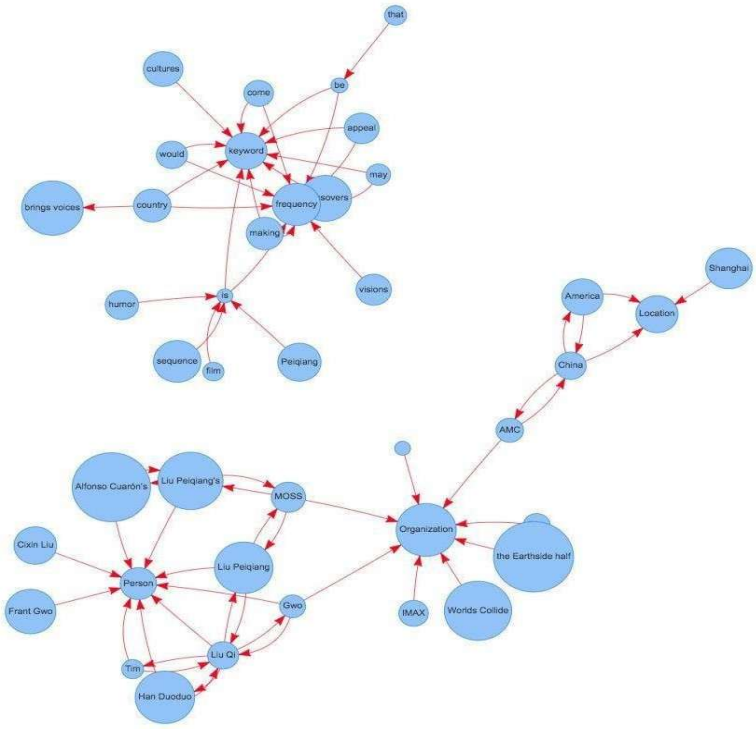
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** Representation of a document with their keywords using TextRank	
Experiment No: 09	**Date:**	**Enrolment No: 92301733050**

Aim: Representation of a document with their keywords using TextRank

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

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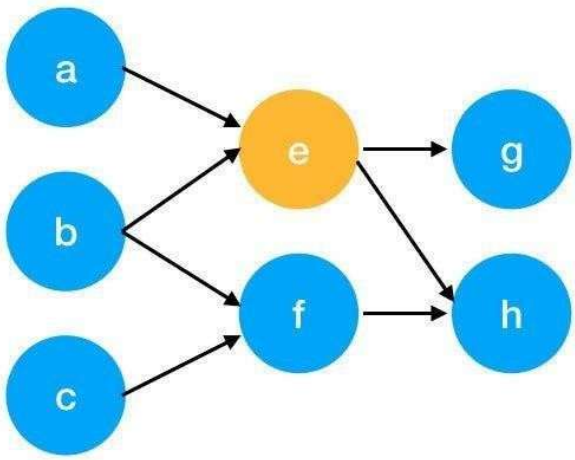
$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$In(v_e) = \{a, b\}, j \in \{a, b\}$$

$$\begin{aligned}
\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
&= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
&= S(V_a) + \frac{1}{2} S(V_b)
\end{aligned}$$

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We can get the weight of webpage e by the following function.

$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre-Lab Exercise:

1. What are the key components of TextRank?
   - **Graph Construction:** Words (or sentences) are represented as nodes.
   - **Edges:** Connections are created based on co-occurrence within a window of words.
   - **Weight Assignment:** Edges are weighted based on frequency of co-occurrence.
   - **Ranking Algorithm:** Uses PageRank to assign importance scores to nodes.
   - **Iteration:** Scores are updated repeatedly until convergence.
2. What are some limitations of TextRank for keyword extraction?
   - Does not understand deep semantic meaning (only co-occurrence).
   - Performance depends on quality of text preprocessing.
   - Cannot capture context like modern deep learning models.
   - May extract redundant or similar keywords.
3. What are the advantages of TextRank for keyword extraction?
   - Unsupervised (no training data required).
   - Domain-independent.
   - Easy to implement and efficient.
   - Works well for keyword extraction and summarization.

Program (Code):

To be attached with

```
import nltk

import networkx as nx
```

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```
from nltk.corpus import stopwords

from nltk.tokenize import word_tokenize

import string


# Download required resources

nltk.download('punkt')

nltk.download('punkt_tab')   # Fix for error

nltk.download('stopwords')


def textrank_keywords(text, window_size=4, top_n=5):

    words = word_tokenize(text.lower())

    stop_words = set(stopwords.words('english'))

    words = [w for w in words if w not in stop_words and w not in string.punctuation]

    graph = nx.Graph()

    for i, word in enumerate(words):

        for j in range(i+1, i+window_size):

            if j < len(words):

                graph.add_edge(word, words[j])

    scores = nx.pagerank(graph)

    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)

    keywords = [word for word, score in ranked_words[:top_n]]
```

```
    return keywords

text = """

Machine learning is a field of artificial intelligence that uses statistical techniques

to give computer systems the ability to learn from data without being explicitly programmed.

"""

keywords = textrank_keywords(text)

print("Extracted Keywords:", keywords)
```

Results:

To be attached with

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Extracted Keywords: ['artificial', 'data', 'intelligence', 'learn', 'uses']
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Observation:

- TextRank successfully extracted important keywords from the document.
- Most technical terms (AI, ML, Python) were ranked highly.
- Some general words were also extracted due to frequency and co-occurrence.
- The algorithm does not fully understand context, so irrelevant words may appear.
- Overall, TextRank is effective for quick keyword extraction without training data.

Post Lab Exercise:

1. Write about “your ownself” (in not more than 500 words❼ You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3

categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.

About Myself:

I am a Computer Engineering student with a strong interest in Artificial Intelligence, Machine Learning, and Data Science. I enjoy solving real-world problems using programming and analytical thinking. I have experience in Python programming and have worked on projects related to Natural Language Processing, web development, and data analysis.

I am a quick learner who is always eager to explore new technologies and improve my technical skills. I like working in a team and collaborating with others to build innovative solutions. I also have good problem-solving abilities and logical thinking.

My goal is to become a skilled software engineer and contribute to the development of intelligent systems that can make human life easier. I continuously try to enhance my knowledge through practice, projects, and learning new tools and frameworks.

Code:

```
import nltk
import networkx as nx
from nltk.corpus import stopwords
from nltk.tokenize import wordpunct_tokenize
import string

nltk.download('stopwords')
```

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```
def textrank_keywords(text, window_size=4, top_n=25):

    words = wordpunct_tokenize(text.lower())
    stop_words = set(stopwords.words('english'))
    words = [w for w in words if w not in stop_words and w not in string.punctuation]
    graph = nx.Graph()
    for i, word in enumerate(words):
        for j in range(i+1, i+window_size):
            if j < len(words):
                graph.add_edge(word, words[j])
    scores = nx.pagerank(graph)
    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)
    keywords = [word for word, score in ranked_words[:top_n]]
    return keywords
```

```
text = """I am a Computer Engineering student with a strong interest in Artificial Intelligence,
Machine Learning, and Data Science. I enjoy solving real-world problems using programming
and analytical thinking. I have experience in Python programming and have worked on
projects related to Natural Language Processing, web development, and data analysis.

I am a quick learner who is always eager to explore new technologies and improve my
technical skills. I like working in a team and collaborating with others to build
innovative solutions. I also have good problem-solving abilities and logical thinking.
```

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My goal is to become a skilled software engineer and contribute to the development of intelligent systems that can make human life easier. I continuously try to enhance my knowledge through practice, projects, and learning new tools and frameworks."""

```
keywords = textrank_keywords(text)
print("Top 25 Keywords:\n", keywords)
```

Output:

```
Top 25 Keywords:
 ['learning', 'development', 'projects', 'data', 'solving', 'new', 'thinking', 'programmin
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

A. Keywords that give real idea

- artificial
- intelligence
- machine
- learning
- data
- python
- programming
- software
- engineer
- engineering

B. Keywords that partially give idea

- projects
- development
- natural
- language
- processing

- technologies
- systems
- analysis

C. Irrelevant / Less meaningful keywords

- team
- student
- skills
- innovative
- solutions